

Quantum Gravitation

Lecture 11

Bounds on Medium Quanta

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$q = (d|s)/\sqrt{(s|s)}$$

■ Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$p(m_g, g|D) \propto L_{prop} L_{scat} L_{therm} p_0$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$m_g^{95} < m_{32}$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: M32-r6 release note; Upper-Limit Registry Manual; Seo, ch. 11